Statement of the problem of creating an application for finding a sea vessel's course

Aim

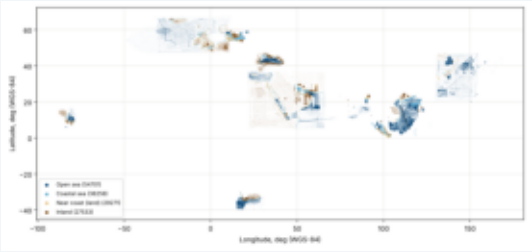
Develop a cross-platform desktop application that searches for the optimal course of a sea vessel over open-sea points with a genetic algorithm and compares it with Dijkstra and A*.

Input: labelled dataset (WGS-84 points, 4 classes), departure and destination, parameters of the graph, penalties for dangerous zones and parameters of the genetic algorithm.
Output: route of the GA, optimal routes of Dijkstra and A*, convergence plot, comparison table, files (PNG, GeoJSON, CSV, parameter log).

Dataset of the student (after cleaning)

Class	Meaning	Points	Role
OPEN_SEA	open sea	54 701	node
COASTAL_SEA	coastal (shallow) sea	36 258	node, penalised
NEAR_COAST	land within 5 km of shore	29 271	obstacle
COASTLINE	inland	27 533	obstacle

152 838 points read 5 075 removed
(class contradicts the measured depth, tolerance 5 m)



Requirements of the assignment

– Windows 10/11 and Linux (Ubuntu 20.04+, Fedora 38+); graphical user interface;
– loading of the dataset from a file (WGS-84, 4 classes of points);
– graph over the points: by distance or by visibility;
– genetic algorithm for the ship's course; avoidance of dangerous zones, minimisation of the path length;
– settings: population size, generations, crossover and mutation probabilities, penalties for dangerous zones;
– Dijkstra and A* for comparison on the same graph;
– map with the points and the three routes, convergence plot of the GA;
– export: map image, routes, convergence plot, parameter log.

Criterion: route cost $F(P) = L(P) + \alpha \cdot Lc(P) + \beta \cdot Ld(P) \rightarrow \min$
L – length, Lc – length in coastal water, Ld – danger-weighted length near land

Rev.	Sheet	Doc. No.	Sign.	Date
Developed		Li H.		
Checked		Kozinsky A.A.		
T. contr.				
N. contr.				
Approved		Golovko V.A.		

КР.ИИ2025.250978-01

Development of an application for finding the optimal course of a sea vessel over open-sea points based on a genetic algorithm

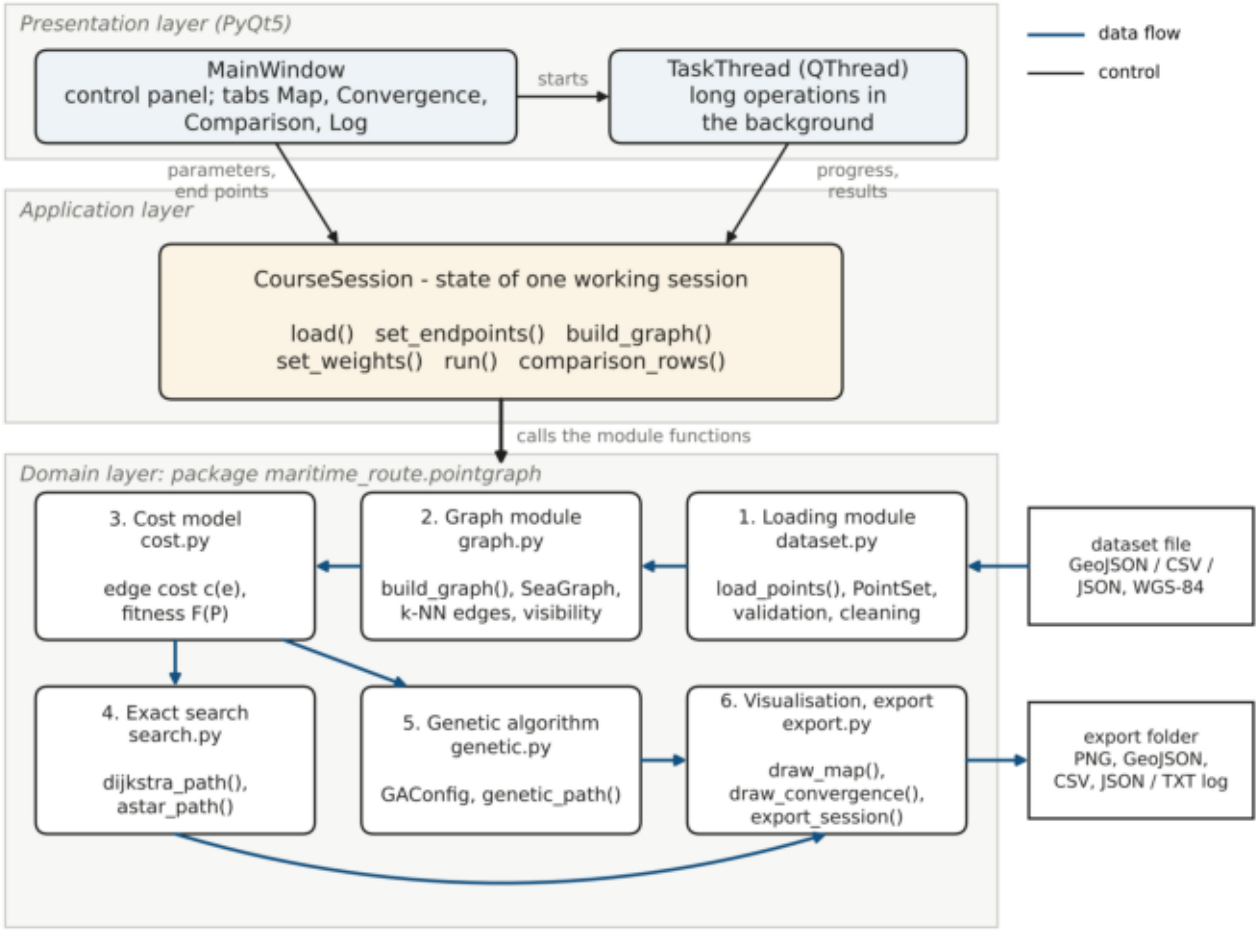
Problem statement (poster)

Lit.	Mass	Scale
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Architecture of the application for finding a sea vessel's course



Module	File	Responsibility
Loading	dataset.py	reading, validation, cleaning
Graph	graph.py	k-NN edges, visibility test
Cost model	cost.py	edge cost, fitness F(P)
Exact search	search.py	Dijkstra, A*
Genetic algorithm	genetic.py	encoding, operators, elitism
Session	session.py	state, presets, comparison
Visualisation	export.py	map, convergence, export
GUI	desktop/*.py	PyQt5 window, background threads

Tools: Python 3.10+, NumPy, SciPy (k-d tree), NetworkX, Matplotlib, PyQt5, folium (optional)

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Architecture (poster)

Lit.	Mass	Scale
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Formalisation of the problem: chromosome and fitness function

Navigation graph and fitness function

G = (V, E): V – water points (OPEN_SEA, COASTAL_SEA) of the area;

E – k nearest neighbours closer than l_max, segments that cross land are removed (visibility test).

Edge cost: $c(e) = l(e) \cdot (1 + \alpha \cdot s(e) + \beta \cdot \varphi(e))$, $\varphi(e) = \max(0, 1 - d_land(e) / r)$

Fitness: $F(P) = \Sigma\, c(e) = L(P) + \alpha \cdot Lc(P) + \beta \cdot Ld(P) \rightarrow \min$

Quality: $\delta = (F_GA - F^*) / F^* \cdot 100\ \%$, F^* – optimum of Dijkstra / A*

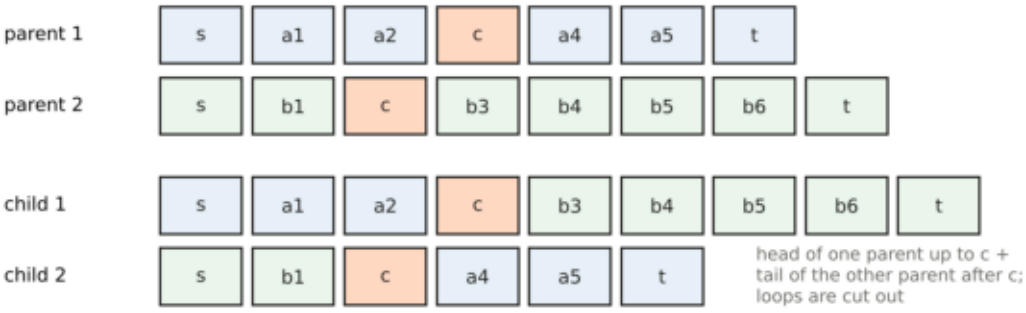
l(e) – great-circle length (haversine), s(e) – share in coastal water, d_land(e) – distance to land,

α, β – penalties, r – radius of the danger zone. $c(e) \geq l(e)$, so $h = d_gc$ is admissible for A*.

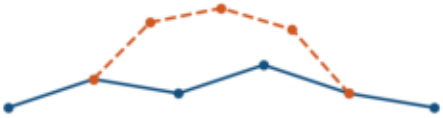
a) chromosome = route of variable length

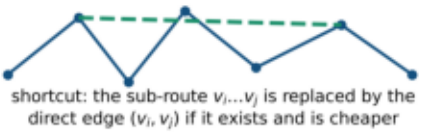


b) one-point crossover at a common node c



c) mutations





Initial population: random walks s → t, $p(v|u) \sim \exp(\kappa \cdot \gamma)$, $\kappa \in [1; 15]$, 50 % through an intermediate node

Selection: tournament (k = 3); crossover: one point at a common node, $p_c = 0.8$

Mutation ($p_m = 0.3$): re-routing (60 %) or shortcut (40 %); elitism: e = 2

Stop: $G_{max} = 300$ generations, 80 generations without improvement or user request

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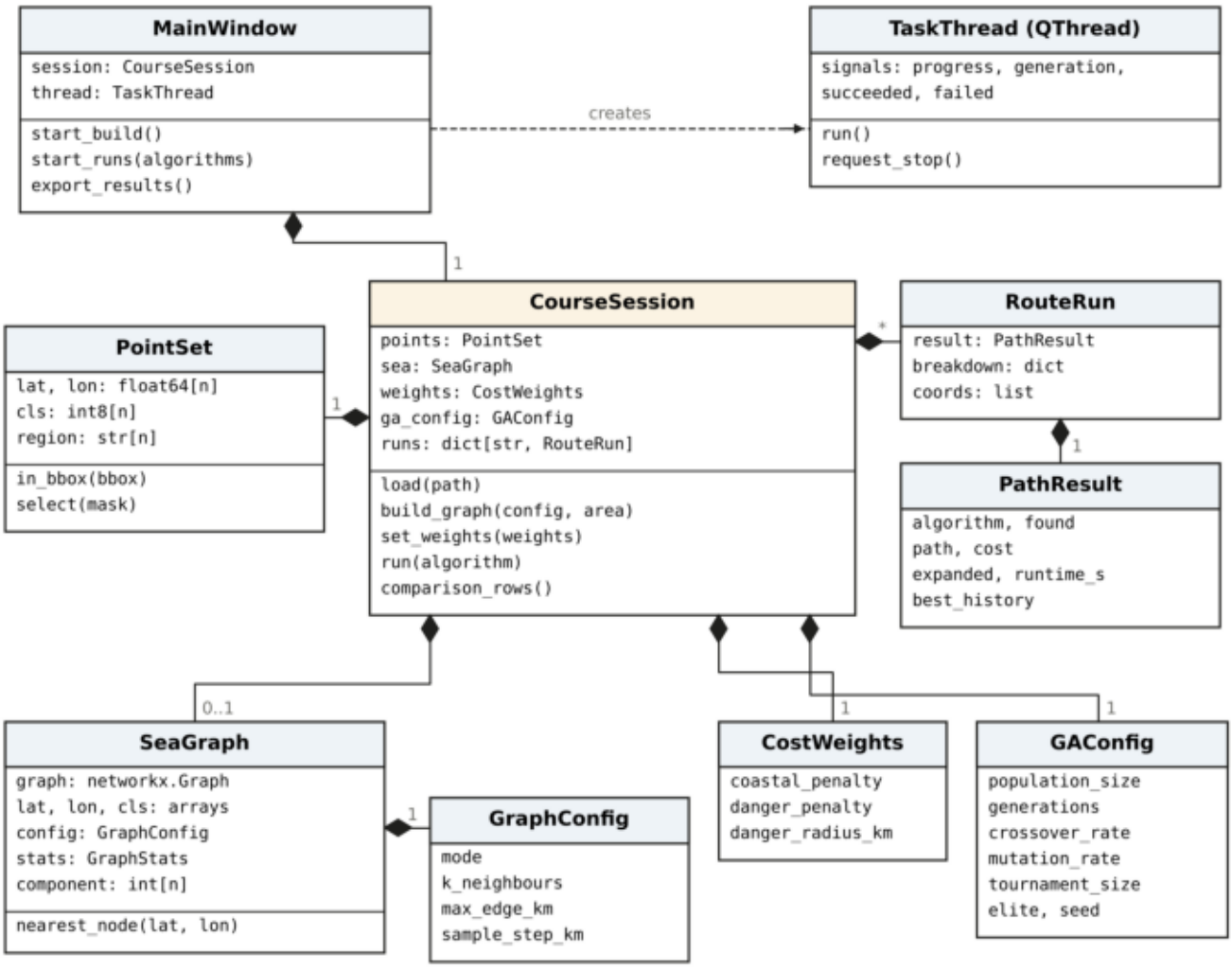
Formalisation (poster)

Lit.	Mass	Scale
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Structural diagram of the application



Filled diamond - composition (the owner keeps the object); dashed arrow - creation.

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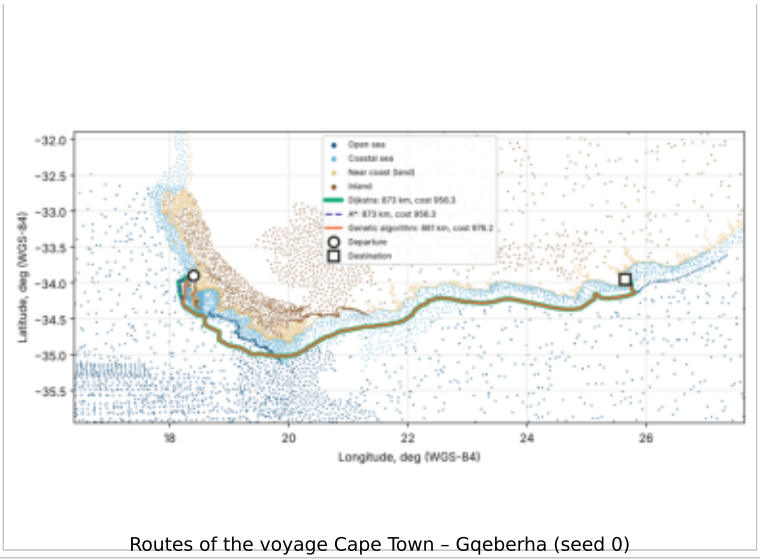
Structural diagram (drawing)

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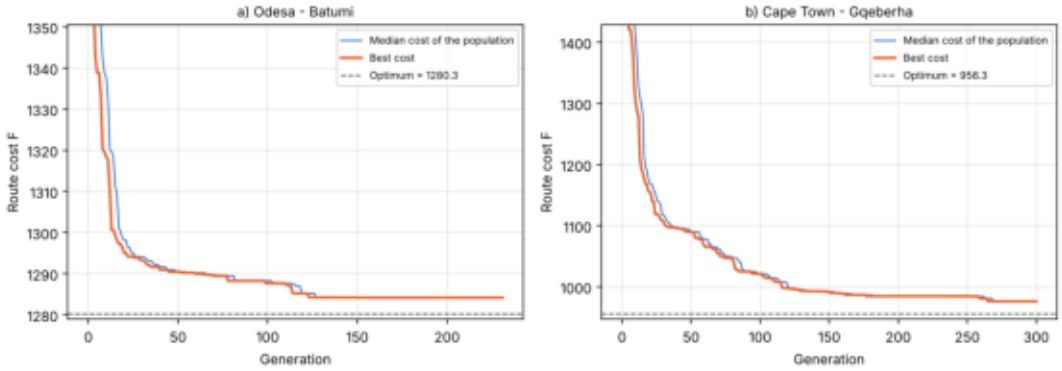
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Test results of the application



Routes of the voyage Cape Town – Gqeberha (seed 0)



Convergence of the GA: Odesa – Batumi, Cape Town – Gqeberha

Voyage	F*	Exp. Dijkstra / A*	Time D / A*, ms	GA mean gap, %	GA ≤ 1 %	GA time, s
Odesa – Batumi	1280.3	5940 / 3465	31 / 27	0.28	10 of 10	4.2
Varna – Sinop	836.4	4478 / 2483	18 / 15	2.07	7 of 10	2.2
Burgas – Trabzon	1220.2	5155 / 4113	24 / 31	0.16	10 of 10	3.1
Sanya – Muara	1636.0	8688 / 1926	48 / 13	0.28	10 of 10	3.1
Cape Town – Gqeberha	956.3	8007 / 7279	38 / 49	2.85	0 of 10	5.7
Dubai – Muscat	694.6	2804 / 1289	12 / 8	0.28	10 of 10	3.3
Colon – Limon	626.0	2319 / 1885	10 / 11	0.37	10 of 10	2.5

– Dijkstra and A* give the same optimum; A* expands on average 1.97 times fewer nodes;

– the GA found valid routes in all 70 runs; on 5 of 7 voyages every run was within 1 % of the optimum;

– on the graph built by distance the optimal route crossed land on 2 of 7 voyages;

– without intermediate nodes in the random walks the mean gap on Cape Town – Gqeberha grows from 2.63 % to 12.50 %.

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Test results (drawing)

Lit.	Mass	Scale
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Sheet 5	Sheets 5

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